

SPSync Multi-Server Scenario (plan 4.7)

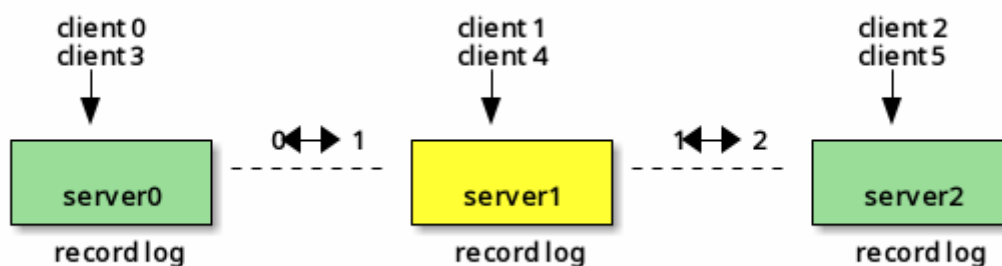
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This is a walk-through of the `multi server full scenario` test (`spsync/engine/test/test_multi_server.cpp`): three server replicas, six clients (two per server), exercising every record kind and event and asserting the replicas converge. The record tags below are from one real run of the test.

Topology

Three servers in a **chain**: server 0 and server 2 are not linked directly, so everything between them must relay through server 1. Two clients sit on each server.



Each server keeps its own record log. A client is synced with the records of the server it is attached to; a server never decrypts (it is content-blind).

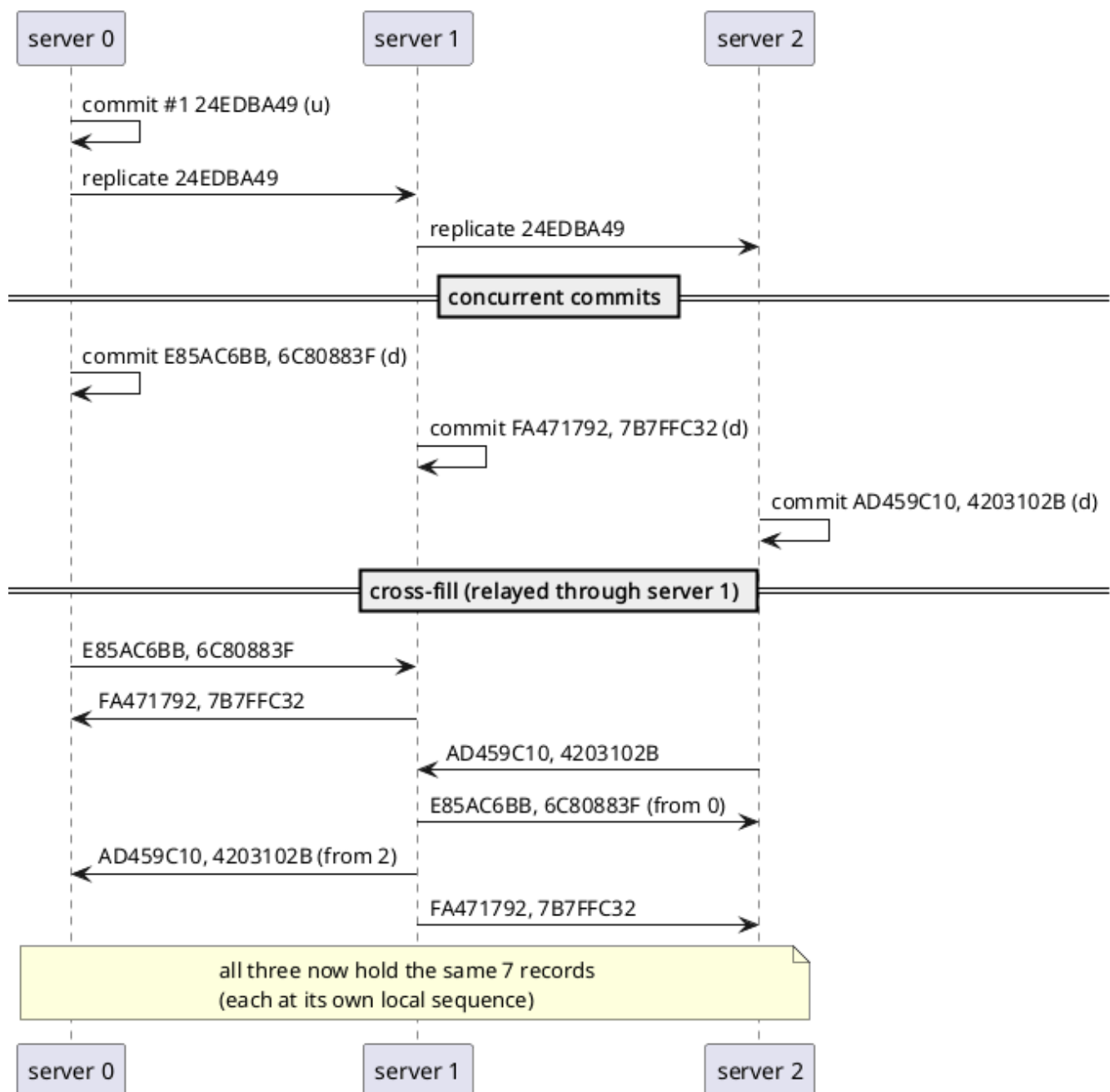
Records and events

Kind	Meaning
(u) user change	membership + encryption-key delivery (D9 merge)
(d) data change	an application object change (a "message")

Kind	Meaning
(s) segment	a signed checkpoint sealing the history before it
commit	a record created locally by a client on that server
replicate	a record received from a peer server (anti-entropy / push)

Phase 1: a change spreads across the chain

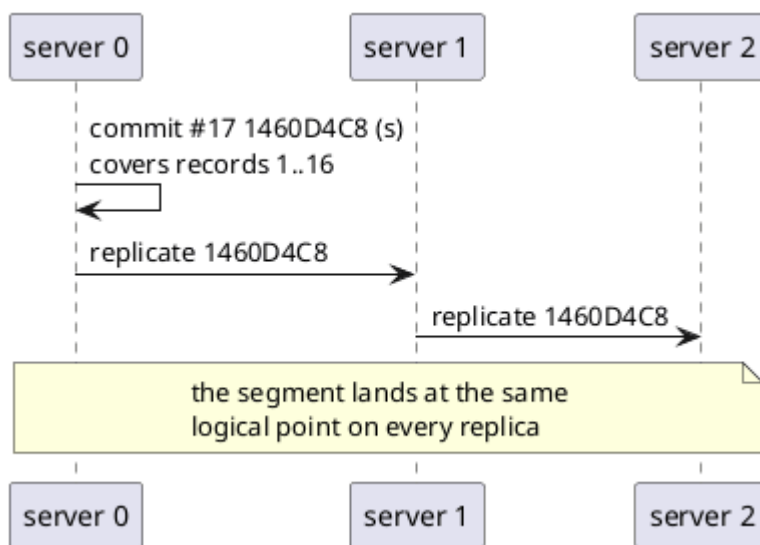
Server 0 commits the first user change; it relays $0 \rightarrow 1 \rightarrow 2$. Then all three servers commit concurrently and the records cross-fill. Note how server 2 receives server 0's records **from server 1** — the transitive relay through the middle.



Phases 2-5: membership, key rotation, segment

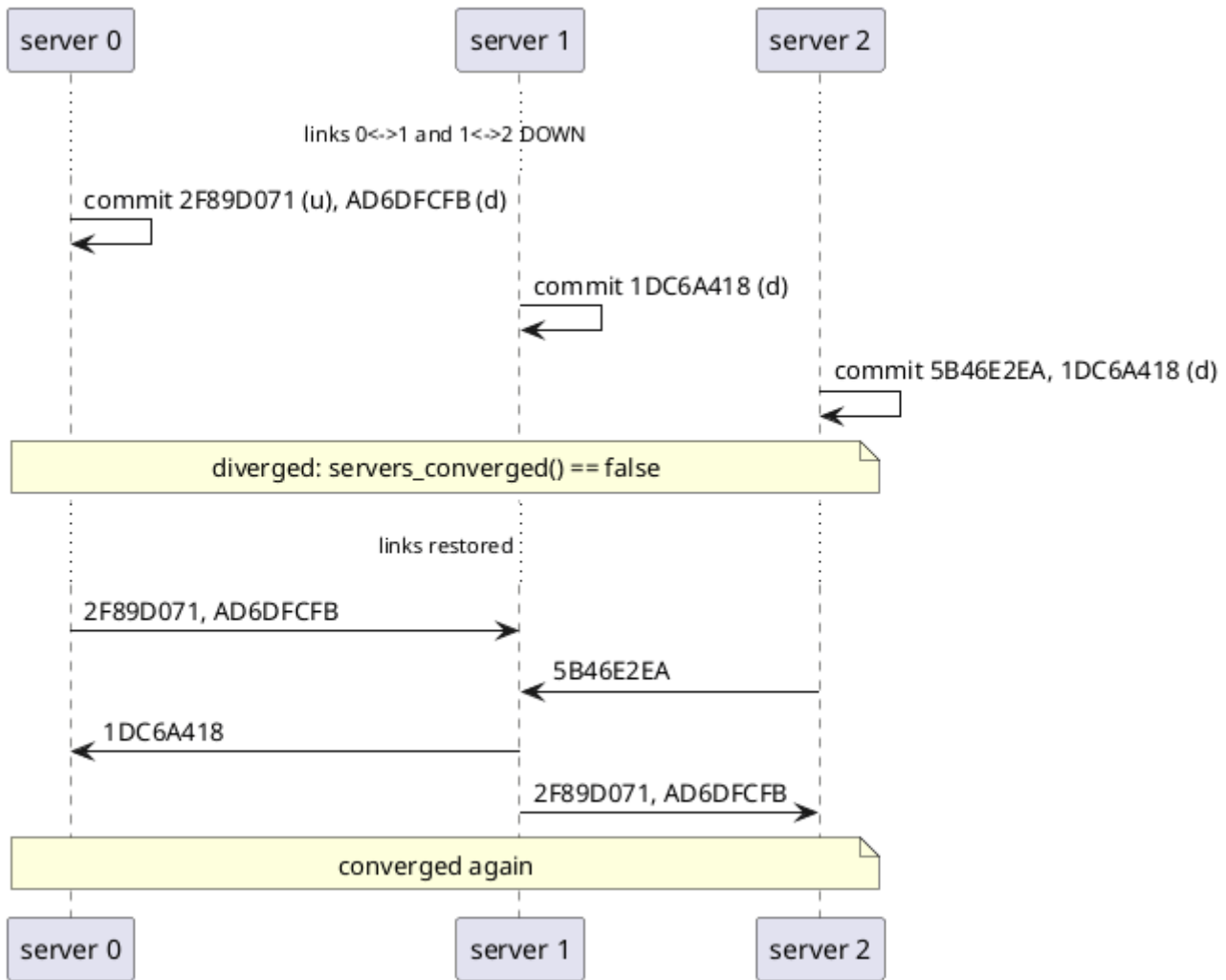
Committed in turn, each draining and asserting convergence:

- **membership** — a user change removes a member (re-affirmed later by a rotation).
- **key rotation** — a client creates a fresh key and hands it to the members inside the user-change KEM envelope; data committed under the new key stays readable.
- **concurrent rotation** — two clients rotate at once, leaving two keys at one key sequence (D9); a record is decrypted by trying each candidate key.
- **segment** — client seals a checkpoint after flushing its pending commits:



Phase 6: partition and heal

Both links go down, isolating server 1. Each side keeps committing and the replicas **diverge** (the test asserts this). When the links come back the missing records flow both ways and they converge again.



A client on the isolated server 1 also **hops** to server 2 during the partition (the 4.5 replica resync): it demotes its records to pending, re-syncs on the new replica under its cursor, and keeps committing there.

Final situation: converged, but observed in different orders (D4)

All three servers end with the **same 45 records, same 6 members, same 32 messages** — but each linearised the set in its own local order. The first records of each server's log (**u** = user change, **d** = data change, **s** = segment):

Server	Local order (first 16 of 45)					
0	24EDBA49(u)	E85AC6BB(d)	6C80883F(d)	FA471792(d)	7B7FFC32(d)	AD459C10(d)
	4203102B(d)	B25E45DF(u)	6D076F62(u)	3D1F183A(d)	8DD15E60(d)	28FC8070(d)
	9CC4CE81(u)	79B1E9D4(d)	689C7BF9(u)	255471B1(d)		
1	24EDBA49(u)	FA471792(d)	7B7FFC32(d)	E85AC6BB(d)	6C80883F(d)	AD459C10(d)
	4203102B(d)	B25E45DF(u)	6D076F62(u)	3D1F183A(d)	8DD15E60(d)	28FC8070(d)
	9CC4CE81(u)	79B1E9D4(d)	689C7BF9(u)	255471B1(d)		
2	24EDBA49(u)	AD459C10(d)	4203102B(d)	FA471792(d)	7B7FFC32(d)	E85AC6BB(d)
	6C80883F(d)	B25E45DF(u)	6D076F62(u)	28FC8070(d)	3D1F183A(d)	8DD15E60(d)
	689C7BF9(u)	255471B1(d)	9CC4CE81(u)	79B1E9D4(d)		

Each server puts **its own** commits first and the peers' after, in arrival order — so the seven phase-1 records appear in three different orders, yet the set is identical and the segment **1460D4C8(s)** sits at record 17 on every replica. This is decision D4: sequence numbers are a per-server cursor, identity is the record tag, and convergence is the same tag set, not the same order.

Property	server 0	server 1	server 2
records	45	45	45
ops	45	45	45
messages	32	32	32
members	6	6	6
converged	✓	✓	✓

The raw run log

The full narration, exactly as the test prints it with **-s** (or on failure):

```
link  0 <-> 1 created
link  1 <-> 2 created
[server 0] commit    #1  user change tag 24EDBA49
[server 1] replicate #1  user change tag 24EDBA49 (from server 0)
[server 2] replicate #1  user change tag 24EDBA49 (from server 1)
[server 0] commit    #2  data change tag E85AC6BB
[server 0] commit    #3  data change tag 6C80883F
[server 0] replicate #4  data change tag FA471792 (from server 1)
[server 0] replicate #5  data change tag 7B7FFC32 (from server 1)
[server 1] commit    #2  data change tag FA471792
[server 1] commit    #3  data change tag 7B7FFC32
[server 1] replicate #4  data change tag E85AC6BB (from server 0)
[server 1] replicate #5  data change tag 6C80883F (from server 0)
[server 1] replicate #6  data change tag AD459C10 (from server 2)
[server 1] replicate #7  data change tag 4203102B (from server 2)
[server 2] commit    #2  data change tag AD459C10
[server 2] commit    #3  data change tag 4203102B
[server 2] replicate #4  data change tag FA471792 (from server 1)
[server 2] replicate #5  data change tag 7B7FFC32 (from server 1)
[server 2] replicate #6  data change tag E85AC6BB (from server 1)
[server 2] replicate #7  data change tag 6C80883F (from server 1)
[server 0] replicate #6  data change tag AD459C10 (from server 1)
[server 0] replicate #7  data change tag 4203102B (from server 1)
[server 0] commit    #8  user change tag B25E45DF
[server 1] replicate #8  user change tag B25E45DF (from server 0)
[server 2] replicate #8  user change tag B25E45DF (from server 1)
[server 0] replicate #9  user change tag 6D076F62 (from server 1)
[server 1] commit    #9  user change tag 6D076F62
[server 2] replicate #9  user change tag 6D076F62 (from server 1)
[server 0] commit    #10 data change tag 8DD15E60
```

```

[server 0] replicate #11 data change tag 3D1F183A (from server 1)
[server 1] commit #10 data change tag 3D1F183A
[server 1] replicate #11 data change tag 8DD15E60 (from server 0)
[server 1] replicate #12 data change tag 28FC8070 (from server 2)
[server 2] commit #10 data change tag 28FC8070
[server 2] replicate #11 data change tag 3D1F183A (from server 1)
[server 2] replicate #12 data change tag 8DD15E60 (from server 1)
[server 0] replicate #12 data change tag 28FC8070 (from server 1)
[server 0] commit #13 user change tag 9CC4CE81
[server 0] commit #14 data change tag 79B1E9D4
[server 1] replicate #13 user change tag 9CC4CE81 (from server 0)
[server 1] replicate #14 data change tag 79B1E9D4 (from server 0)
[server 1] replicate #15 user change tag 689C7BF9 (from server 2)
[server 1] replicate #16 data change tag 255471B1 (from server 2)
[server 2] commit #13 user change tag 689C7BF9
[server 2] commit #14 data change tag 255471B1
[server 2] replicate #15 user change tag 9CC4CE81 (from server 1)
[server 2] replicate #16 data change tag 79B1E9D4 (from server 1)
[server 0] replicate #15 user change tag 689C7BF9 (from server 1)
[server 0] replicate #16 data change tag 255471B1 (from server 1)
[server 0] commit #17 segment tag 1460D4C8
[server 1] replicate #17 segment tag 1460D4C8 (from server 0)
[server 2] replicate #17 segment tag 1460D4C8 (from server 1)
[server 0] replicate #18 data change tag D104DC07 (from server 1)
[server 1] commit #18 data change tag D104DC07
[server 2] replicate #18 data change tag D104DC07 (from server 1)
link 0 <-> 1 DOWN (partition)
link 1 <-> 2 DOWN (partition)
[server 0] commit #19 user change tag 2F89D071
[server 0] commit #20 data change tag AD6DFCFB
[server 1] commit #19 data change tag 1DC6A418
[server 2] commit #19 data change tag 5B46E2EA
[server 2] commit #20 data change tag 1DC6A418
link 0 <-> 1 up
link 1 <-> 2 up
[server 0] replicate #21 data change tag 1DC6A418 (from server 1)
[server 1] replicate #20 user change tag 2F89D071 (from server 0)
[server 1] replicate #21 data change tag AD6DFCFB (from server 0)
[server 1] replicate #22 data change tag 5B46E2EA (from server 2)
[server 2] replicate #21 user change tag 2F89D071 (from server 1)
[server 2] replicate #22 data change tag AD6DFCFB (from server 1)
[server 0] replicate #22 data change tag 5B46E2EA (from server 1)
link 1 <-> 2 DOWN (partition)
[server 0] commit #23 user change tag 120B0415
[server 0] replicate #24 user change tag 886755EE (from server 1)
[server 1] commit #23 user change tag 886755EE
[server 1] replicate #24 user change tag 120B0415 (from server 0)
[server 2] commit #23 data change tag FA93FD0F
link 0 <-> 1 DOWN (partition)
link 1 <-> 2 up
[server 0] commit #25 user change tag FF68536E

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[server 0] commit      #26 data change tag E8A6D302
[server 1] commit      #25 data change tag 086B1A14
[server 1] replicate   #26 data change tag FA93FD0F (from server 2)
[server 1] replicate   #27 data change tag 37A21702 (from server 2)
[server 1] replicate   #28 data change tag 77ADF4FF (from server 2)
[server 1] replicate   #29 data change tag F3C58BC7 (from server 2)
[server 2] commit      #24 data change tag 37A21702
[server 2] commit      #25 data change tag 77ADF4FF
[server 2] commit      #26 data change tag F3C58BC7
[server 2] replicate   #27 user change tag 886755EE (from server 1)
[server 2] replicate   #28 user change tag 120B0415 (from server 1)
[server 2] replicate   #29 data change tag 086B1A14 (from server 1)
link 0 <-> 1 up
link 0 <-> 1 DOWN (partition)
[server 0] commit      #27 data change tag ACC424AF
[server 0] commit      #28 data change tag 59DDDD9F
[server 1] replicate   #30 data change tag 8FBC5D95 (from server 2)
[server 2] commit      #30 data change tag 8FBC5D95
link 1 <-> 2 DOWN (partition)
link 1 <-> 2 up
link 0 <-> 1 up
[server 0] commit      #29 data change tag 17691F5F
[server 0] replicate   #30 data change tag 086B1A14 (from server 1)
[server 0] replicate   #31 data change tag FA93FD0F (from server 1)
[server 0] replicate   #32 data change tag 37A21702 (from server 1)
[server 0] replicate   #33 data change tag 77ADF4FF (from server 1)
[server 0] replicate   #34 data change tag F3C58BC7 (from server 1)
[server 0] replicate   #35 data change tag 8FBC5D95 (from server 1)
[server 1] replicate   #31 user change tag FF68536E (from server 0)
[server 1] replicate   #32 data change tag E8A6D302 (from server 0)
[server 1] replicate   #33 data change tag ACC424AF (from server 0)
[server 1] replicate   #34 data change tag 59DDDD9F (from server 0)
[server 1] replicate   #35 data change tag 17691F5F (from server 0)
[server 1] replicate   #36 data change tag 015E19DB (from server 2)
[server 1] replicate   #37 user change tag 3D1974E7 (from server 2)
[server 2] commit      #31 data change tag 015E19DB
[server 2] commit      #32 user change tag 3D1974E7
[server 2] replicate   #33 user change tag FF68536E (from server 1)
[server 2] replicate   #34 data change tag E8A6D302 (from server 1)
[server 2] replicate   #35 data change tag ACC424AF (from server 1)
[server 2] replicate   #36 data change tag 59DDDD9F (from server 1)
[server 2] replicate   #37 data change tag 17691F5F (from server 1)
[server 0] commit      #36 data change tag B3EF69F1
[server 0] commit      #37 user change tag A04281E8
[server 0] replicate   #38 data change tag 015E19DB (from server 1)
[server 0] replicate   #39 user change tag 3D1974E7 (from server 1)
[server 0] replicate   #40 user change tag 58EF1135 (from server 1)
[server 1] commit      #38 user change tag 58EF1135
[server 1] replicate   #39 data change tag B3EF69F1 (from server 0)
[server 1] replicate   #40 user change tag A04281E8 (from server 0)
[server 1] replicate   #41 data change tag 5C7DCB37 (from server 2)

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[server 1] replicate #42 data change tag 90734B07 (from server 2)
[server 2] commit #38 data change tag 5C7DCB37
[server 2] commit #39 data change tag 90734B07
[server 2] replicate #40 user change tag 58EF1135 (from server 1)
[server 2] replicate #41 data change tag B3EF69F1 (from server 1)
[server 2] replicate #42 user change tag A04281E8 (from server 1)
link 1 <-> 2 DOWN (partition)
link 1 <-> 2 up
link 1 <-> 2 DOWN (partition)
link 1 <-> 2 up
[server 0] commit #41 data change tag 7D82D895
[server 0] replicate #42 data change tag 5C7DCB37 (from server 1)
[server 0] replicate #43 data change tag 90734B07 (from server 1)
[server 1] replicate #43 data change tag 7D82D895 (from server 0)
[server 2] replicate #43 data change tag 7D82D895 (from server 1)
[server 0] commit #44 data change tag B262E0C6
[server 0] replicate #45 data change tag 84DE6280 (from server 1)
[server 1] commit #44 data change tag 84DE6280
[server 1] replicate #45 data change tag B262E0C6 (from server 0)
[server 2] replicate #44 data change tag 84DE6280 (from server 1)
[server 2] replicate #45 data change tag B262E0C6 (from server 1)
=== final situation ===
server 0: 45 records, 45 ops, 32 messages, members: [1CDC7A1E, 1E018678, 25F6573F,
67AAD0E4, D99A0565, E97EA431]
order: 24EDBA49(u) E85AC6BB(d) 6C80883F(d) FA471792(d) 7B7FFC32(d) AD459C10(d)
4203102B(d) B25E45DF(u)
6D076F62(u) 8DD15E60(d) 3D1F183A(d) 28FC8070(d) 9CC4CE81(u) 79B1E9D4(d)
689C7BF9(u) 255471B1(d)
1460D4C8(s) D104DC07(d) 2F89D071(u) AD6DFCFB(d) 1DC6A418(d) 5B46E2EA(d)
120B0415(u) 886755EE(u)
FF68536E(u) E8A6D302(d) ACC424AF(d) 59DDDD9F(d) 17691F5F(d) 086B1A14(d)
FA93FD0F(d) 37A21702(d)
77ADF4FF(d) F3C58BC7(d) 8FBC5D95(d) B3EF69F1(d) A04281E8(u) 015E19DB(d)
3D1974E7(u) 58EF1135(u)
7D82D895(d) 5C7DCB37(d) 90734B07(d) B262E0C6(d) 84DE6280(d)
server 1: 45 records, 45 ops, 32 messages, members: [1CDC7A1E, 1E018678, 25F6573F,
67AAD0E4, D99A0565, E97EA431]
order: 24EDBA49(u) FA471792(d) 7B7FFC32(d) E85AC6BB(d) 6C80883F(d) AD459C10(d)
4203102B(d) B25E45DF(u)
6D076F62(u) 3D1F183A(d) 8DD15E60(d) 28FC8070(d) 9CC4CE81(u) 79B1E9D4(d)
689C7BF9(u) 255471B1(d)
1460D4C8(s) D104DC07(d) 1DC6A418(d) 2F89D071(u) AD6DFCFB(d) 5B46E2EA(d)
886755EE(u) 120B0415(u)
086B1A14(d) FA93FD0F(d) 37A21702(d) 77ADF4FF(d) F3C58BC7(d) 8FBC5D95(d)
FF68536E(u) E8A6D302(d)
ACC424AF(d) 59DDDD9F(d) 17691F5F(d) 015E19DB(d) 3D1974E7(u) 58EF1135(u)
B3EF69F1(d) A04281E8(u)
5C7DCB37(d) 90734B07(d) 7D82D895(d) 84DE6280(d) B262E0C6(d)
server 2: 45 records, 45 ops, 32 messages, members: [1CDC7A1E, 1E018678, 25F6573F,
67AAD0E4, D99A0565, E97EA431]
order: 24EDBA49(u) AD459C10(d) 4203102B(d) FA471792(d) 7B7FFC32(d) E85AC6BB(d)

```

```
6C80883F(d) B25E45DF(u)
    6D076F62(u) 28FC8070(d) 3D1F183A(d) 8DD15E60(d) 689C7BF9(u) 255471B1(d)
9CC4CE81(u) 79B1E9D4(d)
    1460D4C8(s) D104DC07(d) 5B46E2EA(d) 1DC6A418(d) 2F89D071(u) AD6DFCFB(d)
FA93FD0F(d) 37A21702(d)
    77ADF4FF(d) F3C58BC7(d) 886755EE(u) 120B0415(u) 086B1A14(d) 8FBC5D95(d)
015E19DB(d) 3D1974E7(u)
    FF68536E(u) E8A6D302(d) ACC424AF(d) 59DDDD9F(d) 17691F5F(d) 5C7DCB37(d)
90734B07(d) 58EF1135(u)
    B3EF69F1(d) A04281E8(u) 7D82D895(d) 84DE6280(d) B262E0C6(d)
client 0 (on server 0): 45 records, 0 pending
    order: same as its server
client 1 (on server 2): 45 records, 0 pending
    order: same as its server
client 2 (on server 2): 45 records, 0 pending
    order: same as its server
client 3 (on server 0): 45 records, 0 pending
    order: same as its server
client 4 (on server 1): 45 records, 0 pending
    order: same as its server
client 5 (on server 2): 45 records, 0 pending
    order: same as its server
converged: YES
=====
```